

Vendor	# Qty	Tr	Part Number from Vendor	Description from Vendor	Price/Part	Extended
Pololu	1		4752	30:1 Metal Gearmotor 37Dx68L 12V with 64 Cl	\$70.95	DC Motor
Pololu	1		1999	Pololu Universal Aluminium Mounting Hub for i	\$12.95	Hubs for mounting wheel to motor
Pololu	1		1435	Pololu Wheel 90x10mm Pair - Black	\$9.49	Wheel for motor
Digikey	1		IRLZ44NPBF-ND	MOSFET N-CH 55V 47A T0220AB	\$1.33	MOSFET to drive motor
Digikey	4		490-2971-ND	TRIMMER 100k OHM 0.25W PC PIN TOP	\$1.90	Potentiometers for Kp, Ki, Triangle Waveform adjustment, F/V Adjust
Digikey	1		1655-15SQ045CT-ND	DIODE SCHOTTKY 45V 15A R-6	\$1.14	Schottky Diode for Circuit Protection
Digikey	6		296-44779-5-ND	IC OPAMP GP 2 CIRCUIT 8SOIC	\$2.74	High Precision Op-Amps
Digikey	1		LM2917M-8/NOPB-ND	IC F/V CONV 10KHZ 8SOIC	\$1.83	F/V Converter
Digikey	1		MCP1407-E/P	IC GATE DRVR LOW-SIDE 8DIP	\$1.25	Gate driver
N/A	1		XR2206CP-F	IC FUNTION GENERATOR 16DIP	\$0.00	Obselete. Acquired from Hacker Space. Part number and description
Digikey	2		1528-1071-ND	SMT ADAP 6 PACK 8SOIC/MSOP/TSSOP	\$2.95	SOIC8 Adapters for breadboard/protoboard
Digikey	1		1866-3359-ND	AC/DC CONVERTER 12V 72W	\$14.10	Power supply
Digikey	1		LM311NNS/NOPB-ND	IC COMPARATOR 1 GEN PUR 8DIP	\$0.92	Comparator
Digikey	1		PTV09A-4015F-B104-ND	POT 100K OHM 1/20W CARBON LINEAR	\$0.89	Potentiometer for setpoint adjustment
HiLetgo	1		N/A	20pcs 40P 2.54mm Pitch Single Row Pin Hea	\$5.49	Header pins for connecting SOIC adapters to protoboard
Mouser	1		512-LM7812CTG	Linear Voltage Regulators REGULATOR 12V 1A	\$0.50	Positive 12V Linear Regulator
Mouser	1		926-LM7912CT/NOPB	Linear Voltage Regulators 3-Terminal Neg Vltg	\$1.48	Negative 12V Linear Regulator
Mouser	1		580-MEA1D1215SC	DC/DC Converters - Through Hole 12VIN +/- 1%	\$5.07	DC-DC Power Supply
Whitney Cabinet	26		N/A	0.1uF capacitors	\$0.00	
Whitney Cabinet	4		N/A	1uF capacitors	\$0.00	
Whitney Cabinet	1		N/A	3.9 nF capacitor	\$0.00	
Whitney Cabinet	2		N/A	2.7nF capacitor	\$0.00	
Whitney Cabinet	1		N/A	1k ohm resistor	\$0.00	
Whitney Cabinet	7		N/A	100k ohm resistors	\$0.00	
Whitney Cabinet	1		N/A	25k ohm resistor	\$0.00	
Whitney Cabinet	1		N/A	470 ohm resistor	\$0.00	
Whitney Cabinet	3		N/A	10k ohm resistor	\$0.00	
Whitney Cabinet	1		N/A	82k ohm resistor	\$0.00	
Whitney Cabinet	1		N/A	56k ohm resistor	\$0.00	
Whitney Cabinet	1		N/A	51k ohm resistor	\$0.00	
Whitney Cabinet	3		N/A	20k ohm resistor	\$0.00	
Whitney Cabinet	2		N/A	5.1k ohm resistor	\$0.00	
Whitney Cabinet	1		N/A	2k ohm resistor	\$0.00	
Whitney Cabinet	1		N/A	3k ohm resistor	\$0.00	
Totals	83				\$ 139.73	